

§1.56(11) projective (source) vs §1.57 choice (target)

§1.56(11): A projective $\iff$ every entire relation from A contains a map

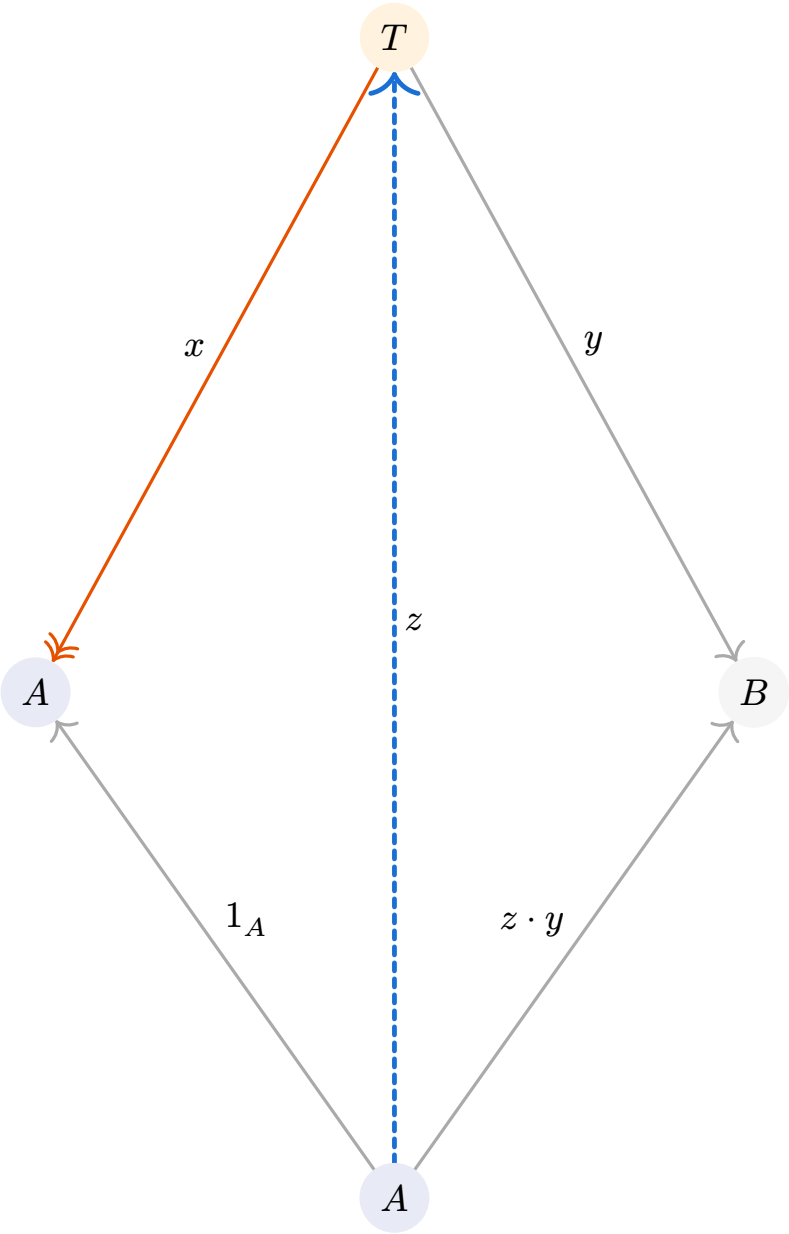


Figure 1: R is entire **from** A. The left leg x is a cover **to** A.
A projective $\implies x$ splits: $z \cdot x = 1_A$.
 z witnesses $\text{graph}(z \cdot y) \subset R$ – the map $z \cdot y : A \rightarrow B$ is contained in R .
The projective object A is the source of the entire relation.

§1.57: B choice $\iff$ every entire relation to B contains a map

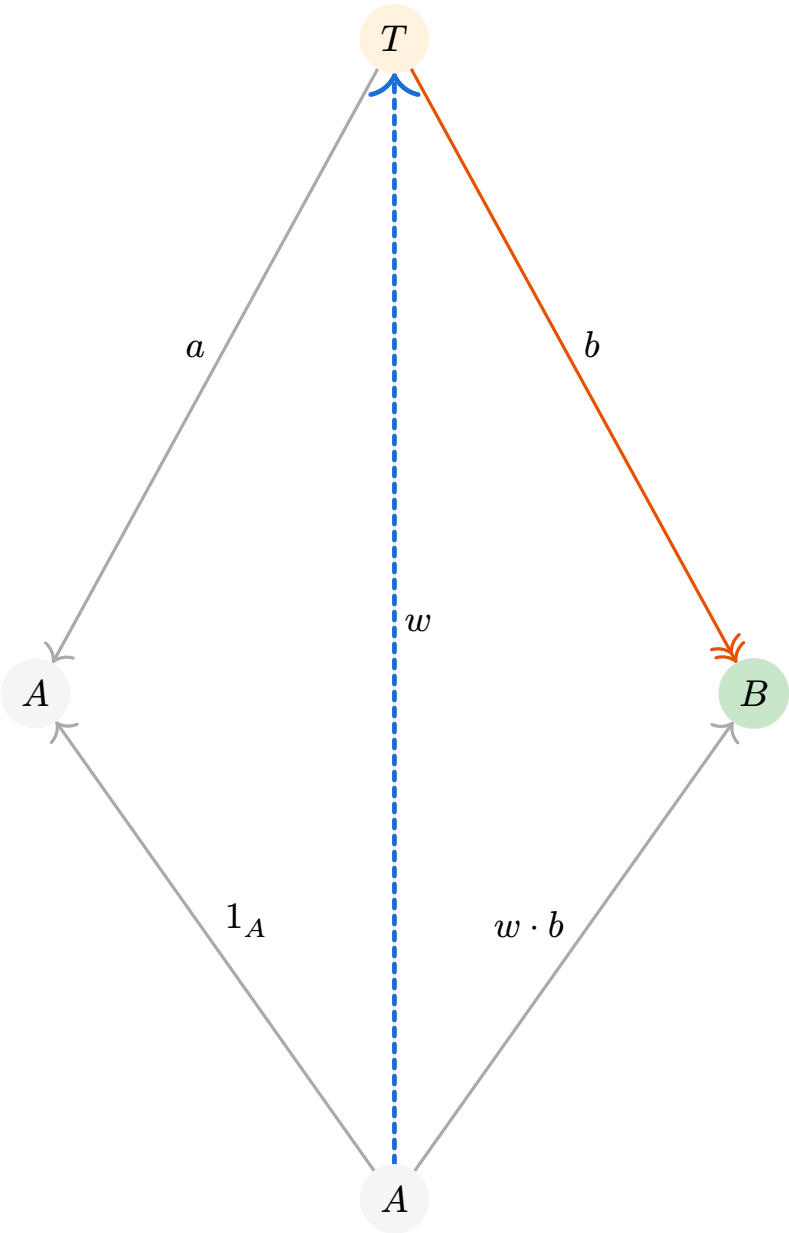


Figure 2: R is entire **to** B. For a relation **targeted at** B, “entire” means the **right** leg is a cover (via reciprocal: R entire to B $\iff R^\circ$ entire from B).
B choice $\implies$ the right leg b splits: $w \cdot b = 1_B$ (but $w : A \rightarrow T$, so the map is w composed with the left leg...).
The choice object B is the target of the entire relation.

Comparison	
§ 1.56(11) projective	§ 1.57 choice
quantifies over source of R	quantifies over target of R
$R : A \rightarrow B$ entire	$R : A \rightarrow C$ entire
left leg x is cover, splits	right leg y is cover, splits (via R°)
map $z \cdot y : A \rightarrow B$	map $z' \cdot x : A \rightarrow C$
dual — same proof, swap columns	